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Certificate of Attendance and Presentation

This certificate recognizes that

JAMAL ATTMANI

(in collaboration with KARAMI FAHD, ATLAS ABDELGHAFOR)

attended the sixth Edition of the **International Conference on Research in Applied Mathematics and Computer Science (ICRAMCS 2024)** and presented a contribution titled:

Convergence Analysis of a Finite Volume Scheme for a Parabolic System



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Convergence Analysis of a Finite Volume Scheme for a Parabolic System

Communication Info

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Abstract

This work analyzes a finite volume scheme for a nonlinear reaction-diffusion system applied in image processing. We first demonstrate the existence of a solution within this scheme. Then, using a series of derived estimates and Kolmogorov's compactness criterion, we rigorously prove the convergence of the finite volume solution to the weak solution of the system. Finally, numerical simulations are presented.

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